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INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N°_7 – Réseaux I

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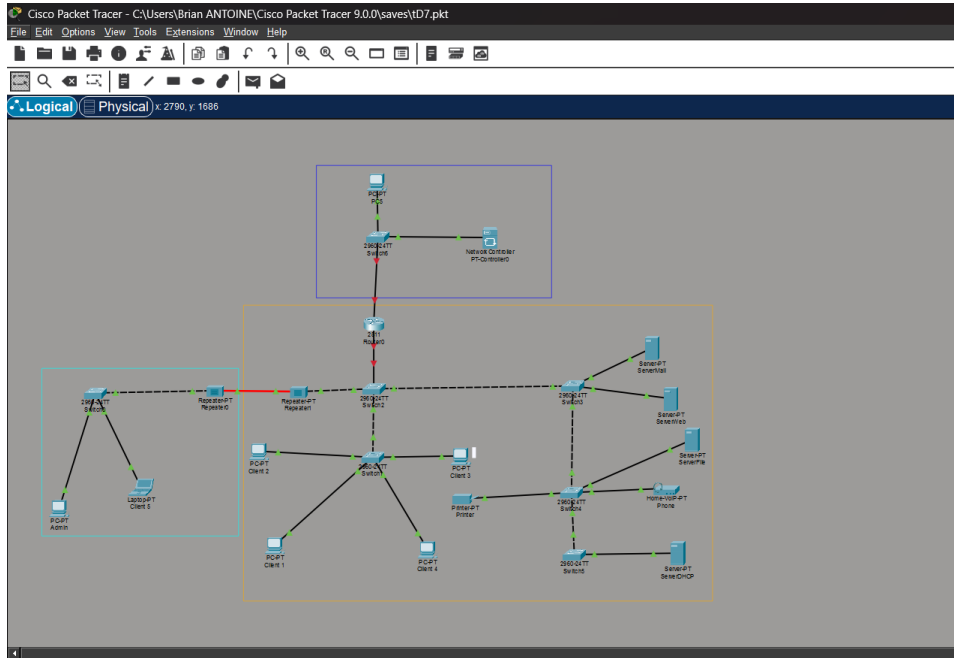
Présenté à: M. Ismaël SAINT AMOUR

Niveau : L3

Date: 13/12/2025

Travaux dirigés

- **Reproduction de la topologie**



A. Configuration Telnet (partie non sécurisée)

➤ Configuration du routeur

The screenshot displays a Cisco IOS command-line interface. At the top, there are tabs for 'Physical', 'Config', 'CLI', and 'Attributes', with 'CLI' currently selected. The main window shows the following text:

```
States and local country laws governing import, export, transfer and
of Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wol/export/crypto/ctoc/statg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249556K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-3-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#
```

At the bottom right of the window, there are two buttons: 'Copy' and 'Paste'.

➤ Configuration PC

The screenshot shows the 'Client 1' configuration window with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Config' tab is active, and the 'IP Configuration' section is expanded. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 192.168.1.2, Subnet Mask: 255.255.255.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The fields are: IPv6 Address (empty), Link Local Address: FE80::205:5EFF:FEA4:821D, Default Gateway (empty), and DNS Server (empty). The '802.1X' section has 'Use 802.1X Security' checked, 'Authentication' set to 'MD5', and 'Username' and 'Password' fields are empty. A 'Top' button is at the bottom left.

1) Configuration du nom, la protection globale et le domaine local

■ Configuration de Telnet

The screenshot shows the 'Router1' configuration window with tabs for Physical, Config, CLI, and Attributes. The 'CLI' tab is active. The text in the CLI window is as follows:

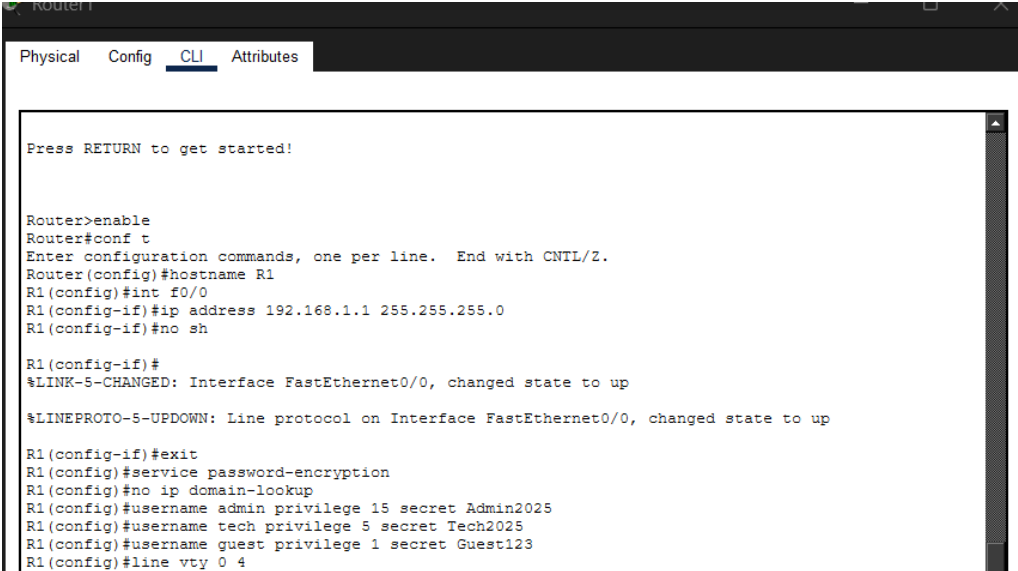
```
Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#service password-encryption
...
```

2) Création des utilisateurs (privilege 1, 5,15)



```
Router1
Physical Config CLI Attributes

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#service password-encryption
R1(config)#no ip domain-lookup
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest123
R1(config)#line vty 0 4
```

3) Activation de Telnet sur les lignes VTY :



```
Router1
Physical Config CLI Attributes

Press RETURN to get started!

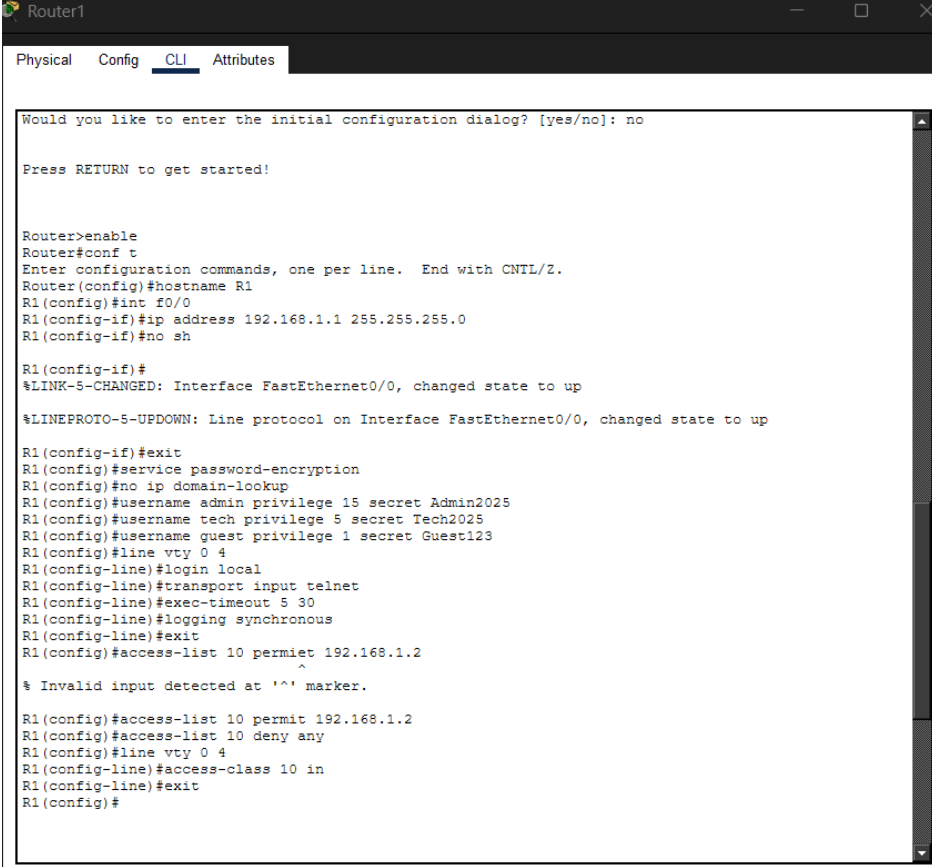
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#service password-encryption
R1(config)#no ip domain-lookup
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest123
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input telnet
R1(config-line)#exec-timeout 5 30
R1(config-line)#logging synchronous
R1(config-line)#exit
```

4) Configuration du mot de passe de ligne :

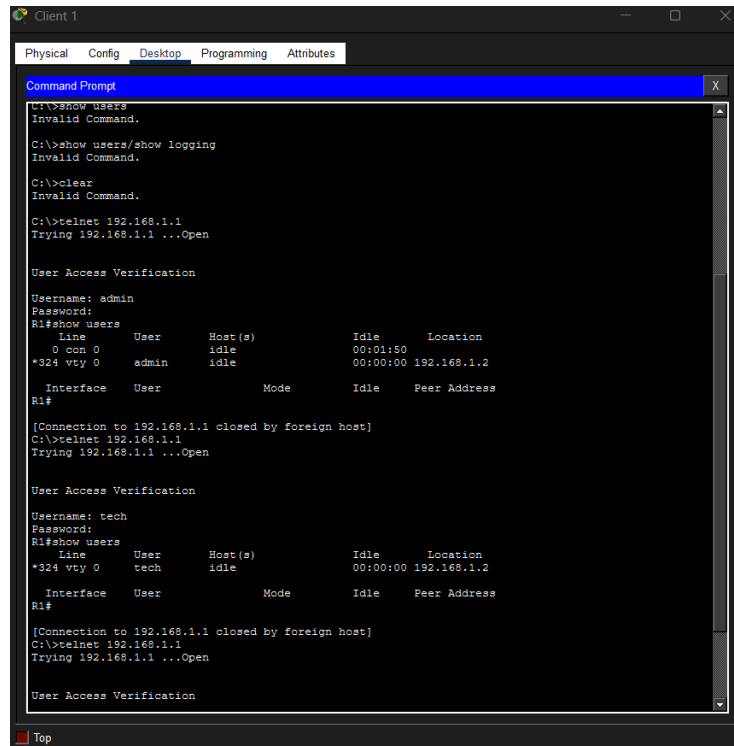


```
Router1
Physical Config CLI Attributes
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh
R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#service password-encryption
R1(config)#no ip domain-lookup
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest123
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input telnet
R1(config-line)#exec-timeout 5 30
R1(config-line)#logging synchronous
R1(config-line)#exit
R1(config)#access-list 10 permit 192.168.1.2
^
% Invalid input detected at '^' marker.
R1(config)#access-list 10 permit 192.168.1.2
R1(config)#access-list 10 deny any
R1(config)#line vty 0 4
R1(config-line)#access-class 10 in
R1(config-line)#exit
R1(config)#
```

Copy Paste

Top

5) Test d'accès Telnet :



The screenshot shows a 'Client 1' window in Cisco Packet Tracer. It contains a 'Command Prompt' window with the following text:

```
C:\>show users
Invalid Command.

C:\>show users/show logging
Invalid Command.

C:\>clear
Invalid Command.

C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification

Username: admin
Password:
R1#show users

   Line    User      Host(s)      Idle    Location
  *324 vty 0   admin      idle        00:01:50  192.168.1.2

   Interface  User      Mode    Idle    Peer Address
R1#

[Connection to 192.168.1.1 closed by foreign host]
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification

Username: tech
Password:
R1#show users

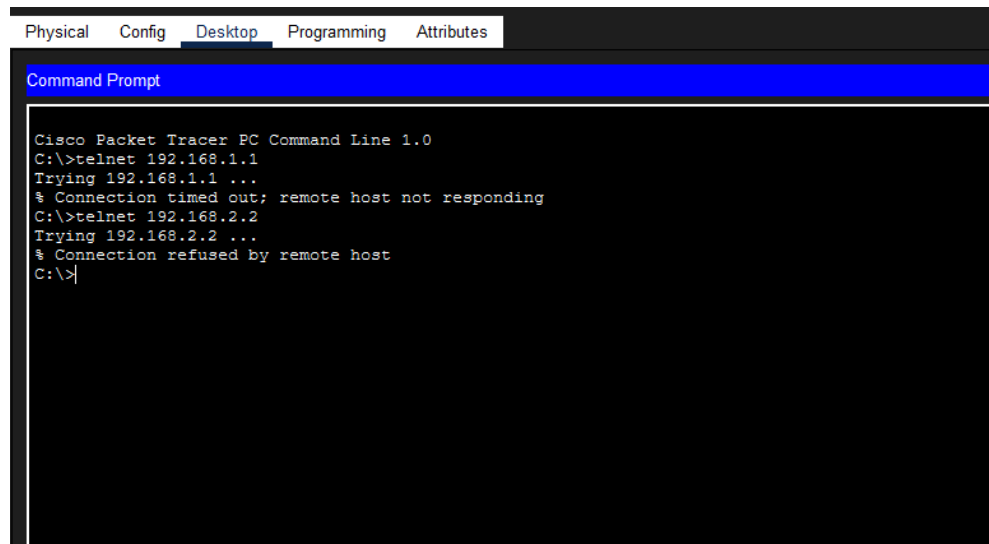
   Line    User      Host(s)      Idle    Location
  *324 vty 0   tech       idle        00:00:00  192.168.1.2

   Interface  User      Mode    Idle    Peer Address
R1#

[Connection to 192.168.1.1 closed by foreign host]
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification
```

PC2 : refuse

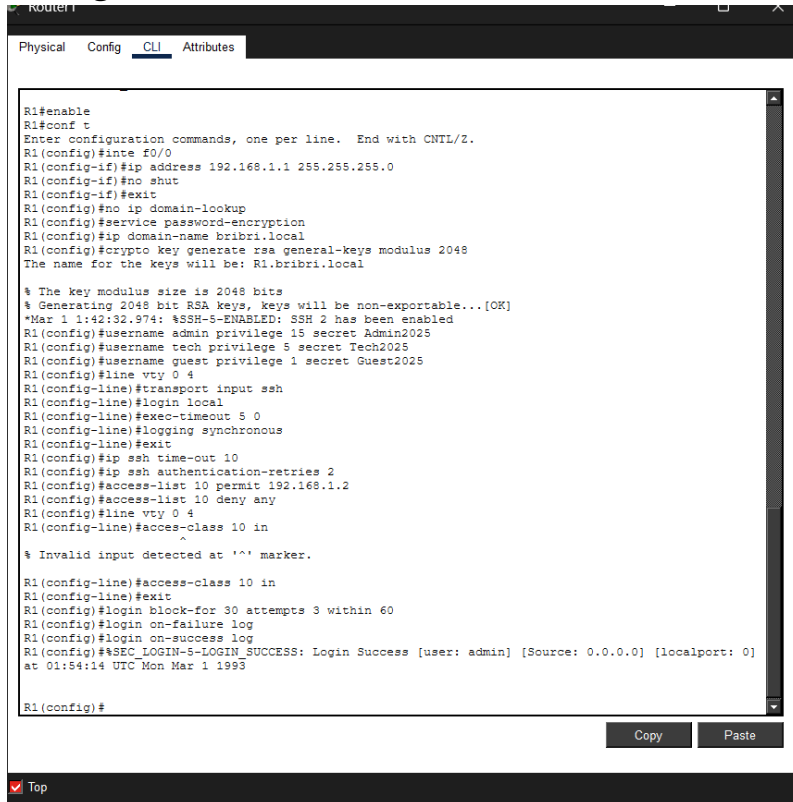


The screenshot shows a 'PC Command Line' window in Cisco Packet Tracer. It contains the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...
% Connection timed out; remote host not responding
C:\>telnet 192.168.2.2
Trying 192.168.2.2 ...
% Connection refused by remote host
C:\>
```

B. Configuration SSH (partie sécurisée)

- Configuration SSH

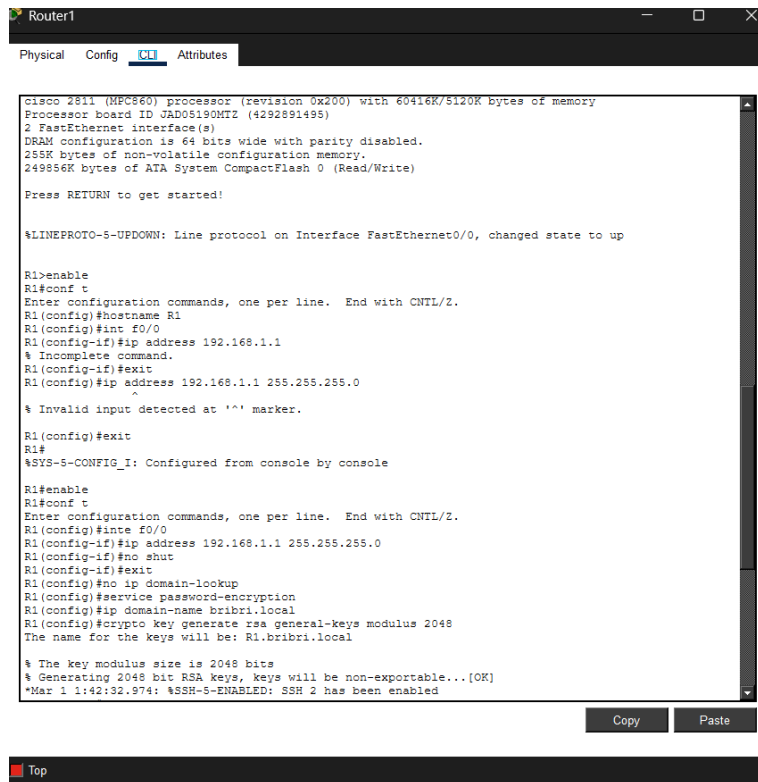


```
R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#inte f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#no ip domain-lookup
R1(config)#service password-encryption
R1(config)#ip domain-name briibri.local
R1(config)#crypto key generate rsa general-keys modulus 2048
The name for the keys will be: R1.briibri.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA Keys, keys will be non-exportable...[OK]
*Mar 1 1:42:32.974: %SSH-5-ENABLED: SSH 2 has been enabled
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest2025
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exec-timeout 5 0
R1(config-line)#logging synchronous
R1(config-line)#exit
R1(config)#ip ssh time-out 10
R1(config)#ip ssh authentication-retries 2
R1(config)#access-list 10 permit 192.168.1.2
R1(config)#access-list 10 deny any
R1(config)#line vty 0 4
R1(config-line)#access-class 10 in
^
% Invalid input detected at '^' marker.
R1(config-line)#access-class 10 in
R1(config-line)#exit
R1(config)#login block-for 30 attempts 3 within 60
R1(config)#login on-failure log
R1(config)#login on-success log
R1(config)#%SEC_LOGIN-5-LOGIN SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0]
at 01:54:14 UTC Mon Mar 1 1993

R1(config)#
```

- Clé de RSA générée (2848 bits)



```
Cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1>enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1
% Incomplete command.
R1(config-if)#exit
R1(config)#ip address 192.168.1.1 255.255.255.0
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#no ip domain-lookup
R1(config)#service password-encryption
R1(config)#ip domain-name bribri.local
R1(config)#crypto key generate rsa general-keys modulus 2048
The name for the keys will be: R1.bribri.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:42:32.974: %SSH-5-ENABLED: SSH 2 has been enabled
```

- Création des utilisateurs privilégiés



```
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1
% Incomplete command.
R1(config-if)#exit
R1(config)#ip address 192.168.1.1 255.255.255.0
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#no ip domain-lookup
R1(config)#service password-encryption
R1(config)#ip domain-name bribri.local
R1(config)#crypto key generate rsa general-keys modulus 2048
The name for the keys will be: R1.bribri.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:42:32.974: %SSH-5-ENABLED: SSH 2 has been enabled
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest2025
```

- Activation de ssh version 2


```
R1(config)#ip ssh version 2
R1(config)#ip ssh time-out 10
R1(config)#ip ssh authentication-retries 2
R1(config)#access-list 10 permit 192.168.1.2
R1(config)#access-list 10 deny any
R1(config)#
```

Copy

Paste

Top

- Désactivation Telnet et gardons uniquement SSH :

```
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exec-timeout 5 0
R1(config-line)#logging synchronous
R1(config-line)#exit
R1(config)#ip ssh time-out 10
R1(config)#ip ssh authentication-retries 2
R1(config)#access-list 10 permit 192.168.1.2
R1(config)#access-list 10 deny any
```

Copy

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Top

- Test de l'accès ssh :

✓ Depuis PC1

```

Client 1
Physical Config Desktop Programming Attributes
Command Prompt
Interface      User          Mode          Idle          Peer Address
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 10 secs; Authentication retries: 2
R1#show crypto key mypubkey rsa
% Key pair was generated at: 1:42:32 UTC March 1 1993
Key name: R1.bribri.local
Storage Device: not specified
Usage: General Purpose Key
Key is not exportable.
Key Data:
00003c57 00007b56 00007e67 0000557a 00000e1a 00004ea1 00002205 00002c5b
00007585 00004076 00000f85 00006921 000044a6 000002fb 00007782 000064c7
0000701c 000079a0 00002bbe 000075db 000050cf 00004400 00000e71 7281
% Key pair was generated at: 1:42:32 UTC March 1 1993
Key name: R1.bribri.local.server
Temporary key
Usage: Encryption Key
Key is not exportable.
Key Data:
00002dde 0000700c 00000be0 000013ba 000022d9 00001766 00005232 00001de6
00003631 00006c0b 00003b7f 00007288 000033ec 00007835 00001e2b 00006f4a
0000209d 00001580 00007772 00003e41 00005894 00005f4b 00002443 7b71
R1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 3 messages logged, xml disabled,
filtering disabled
Monitor logging: disabled
Buffer logging: level debugging, 0 messages logged, xml disabled,
filtering disabled

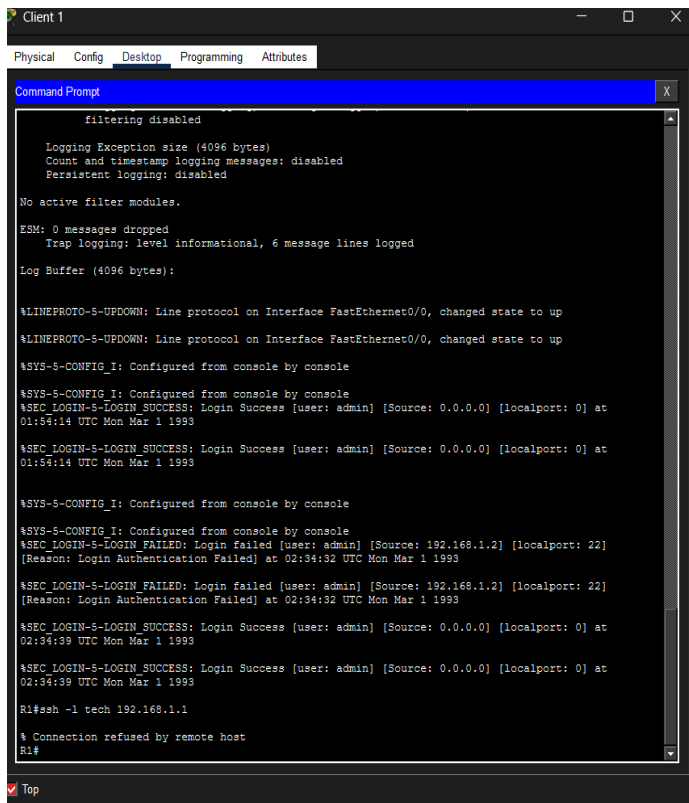
Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 3 message lines logged
  
```

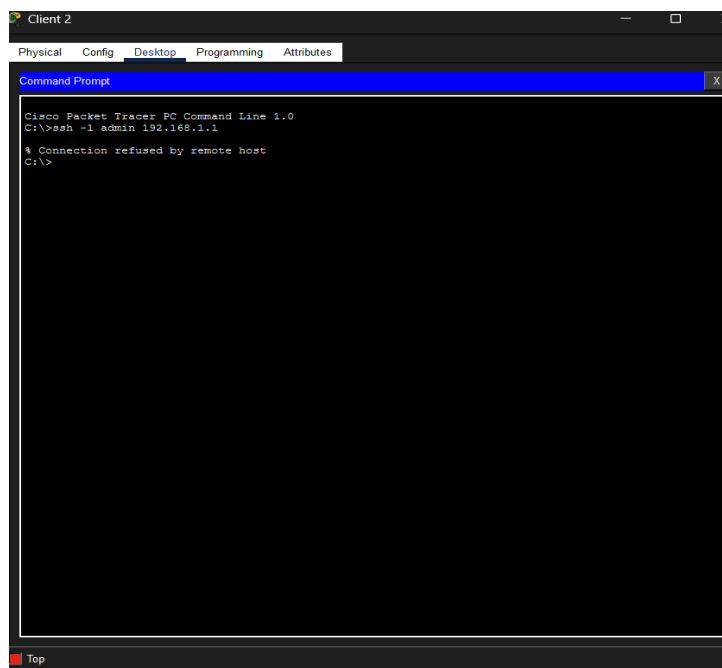
Top

✓ Depuis PC1 en technicien :



```
Client 1
Physical Config Desktop Programming Attributes
Command Prompt
filtering disabled
Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled
No active filter modules.
ESM: 0 messages dropped
Trap logging: level informational, 6 message lines logged
Log Buffer (4096 bytes):
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%SYS-5-CONFIG_I: Configured from console by console
%SYS-5-CONFIG_I: Configured from console by console
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0] at
01:54:14 UTC Mon Mar 1 1993
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0] at
01:54:14 UTC Mon Mar 1 1993
%SYS-5-CONFIG_I: Configured from console by console
%SYS-5-CONFIG_I: Configured from console by console
%SEC_LOGIN-5-LOGIN_FAILED: Login failed [user: admin] [Source: 192.168.1.2] [localport: 22]
[Reason: Login Authentication Failed] at 02:34:32 UTC Mon Mar 1 1993
%SEC_LOGIN-5-LOGIN_FAILED: Login failed [user: admin] [Source: 192.168.1.2] [localport: 22]
[Reason: Login Authentication Failed] at 02:34:32 UTC Mon Mar 1 1993
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0] at
02:34:39 UTC Mon Mar 1 1993
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0] at
02:34:39 UTC Mon Mar 1 1993
R1#ssh -l tech 192.168.1.1
% Connection refused by remote host
R1#
```

○ Depuis PC2 : refusé



```
Client 2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 192.168.1.1
% Connection refused by remote host
C:\>
```